The Dynamic Impedance Phasor Helix:

A Reinterpretation and Combination of Bode & Nyquist Diagrams
for Visualizing the Time Evolution of Impedance
in Power Systems and RLC Circuits

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Abstract

The Bode diagram and the Nyquist diagram are foundational tools in control system analysis, developed in the 1930s and 1940s respectively. Each provides a frequency-domain representation of a system's loop transfer function: Bode as two separate plots of magnitude and phase versus frequency, Nyquist as a single parametric curve in the complex plane. Neither representation explicitly shows how the system evolves in time after a disturbance. This paper presents the *Dynamic Impedance Phasor Helix*, a three-dimensional visualization that unifies the information contained in both classical diagrams by plotting the system's phase portrait—the state variable and its time derivative—extruded along a time axis. The resulting helix encodes magnitude, phase, damping, and temporal evolution in a single geometric object. A stability envelope, derived from a reference system's decay profile, provides a visual boundary for acceptable transient behavior. The paper demonstrates the approach for both power system load frequency control (LFC) and series RLC circuits, showing that the underlying mathematics is identical across domains. The motivation is pedagogical and practical: the author, working as a control systems engineer, found the Nyquist plot difficult to interpret intuitively and sought a representation that would show directly how a system recovers—or fails to recover—after a disturbance.

Keywords: Bode diagram, Nyquist diagram, phasor, impedance, phase portrait, stability visualization, load frequency control, RLC circuit, control system analysis

1 Introduction and Motivation

The Bode diagram, introduced by Hendrik Wade Bode while working at Bell Laboratories in the 1930s [1], represents the frequency response of a linear time-invariant system as two graphs: magnitude (in decibels) versus frequency, and phase (in degrees) versus frequency. The Nyquist diagram, introduced by Harry Nyquist in 1932 [2], maps the same frequency response as a parametric curve in the complex plane, with the stability criterion formulated in terms of encirclement of the critical point $(-1, 0)$.

Both representations are projections of the same underlying object: the complex-valued transfer function $L(j\omega)$ evaluated along the imaginary axis of the s-plane. The Bode diagram decomposes this phasor into its polar components—$|L(j\omega)|$ and $\angle L(j\omega)$—and plots each separately. The Nyquist diagram retains the phasor as a single complex quantity but parameterizes it by frequency, discarding any explicit representation of time.

The author's motivation arose from practical experience in process control and electrical engineering. While the Bode diagram was found to be relatively intuitive—the magnitude plot answers "how much" and the phase plot answers "which direction"—the Nyquist diagram required

significant mental effort to interpret. The need to mentally reconstruct the orthogonal projections of the phasor at each point along the curve, combined with the topological nature of the encirclement criterion, presented a barrier to rapid engineering judgment.

Furthermore, in practical control work, the question of primary interest is not "what are the margins at a particular frequency?" but rather "how will this system behave after a disturbance?" — a question that lives in the time domain. The classical tools answer this question indirectly: the Bode margins predict the qualitative character of the transient response, but do not show it. One must synthesize information from both the Bode and Nyquist representations, along with knowledge of the pole locations, to predict the actual trajectory.

This led to the central question: *What if the information from both diagrams were combined into a single three-dimensional object that explicitly shows the system's evolution in time?*

2 The Phasor as the Fundamental Object

At any single frequency ω, the open-loop transfer function $L(j\omega)$ is a complex number—a phasor. It has a magnitude $|L(j\omega)|$ and a phase $\angle L(j\omega)$. This single phasor contains all the information that the Bode diagram splits into two separate graphs at that frequency.

The phasor can be decomposed into rectangular components:

$$\mathrm{Re}\{L(j\omega)\} \quad \text{— the component parallel to the error (damping axis)} \tag{1}$$

$$\mathrm{Im}\{L(j\omega)\} \quad \text{— the component perpendicular to the error (quadrature axis)} \tag{2}$$

The real part determines whether the feedback correction opposes the error (negative real part, stabilizing) or reinforces it (positive real part, destabilizing). The imaginary part shifts the oscillation frequency but does not add or remove energy. Stability is thus fundamentally determined by the projection of the correction phasor onto the error direction—an inner product.

The Nyquist plot displays both projections directly on its axes. The Bode plot encodes them as polar magnitude and angle. Both are representations of the same phasor. The question is whether there exists a more natural representation—one that retains the full complex structure while adding the dimension that both classical tools discard: time.

3 From Static Phasor to Dynamic Helix

A static phasor represents a steady-state sinusoidal oscillation: constant amplitude, constant phase, rotating at a fixed frequency forever. This is the $j\omega$ axis of the s-plane—the domain of Bode and Nyquist.

A real system responding to a disturbance does not remain at steady state. The oscillation either decays (stable), grows (unstable), or persists (marginally stable). In the complex plane, a decaying oscillation traces a *spiral* that collapses inward toward the origin. A growing oscillation spirals outward. A sustained oscillation traces a circle.

Adding time as a third axis, perpendicular to the complex plane, transforms these spirals into a three-dimensional object: a *helix*. The radius of the helix at each height (time instant) corresponds to the instantaneous magnitude of the phasor—the system's deviation from equilibrium. The rotation corresponds to the oscillation. The rate at which the radius shrinks (or grows) corresponds to the damping (or instability) of the system.

Formally, given a second-order system with dominant poles at $s = \sigma \pm j\omega_d$, the free response has the form:

$$x(t) = A\, e^{\sigma t} \cos(\omega_d t + \phi) \tag{3}$$

In phase-space coordinates $(x, \dot{x})$, this traces a spiral in the plane. Parameterized by time, the trajectory becomes:

$$\mathbf{r}(t) = \big(x(t),\ \dot{x}(t),\ t\big) \in \mathbb{R}^3 \tag{4}$$

This is the Dynamic Phasor Helix. The complex plane $(x, \dot{x})$ forms the base, and time t is the vertical axis. The helix collapses inward when $\sigma < 0$ (stable), maintains constant radius when $\sigma = 0$ (marginally stable), and expands outward when $\sigma > 0$ (unstable).

4 The Stability Envelope

The helix alone shows the trajectory. To assess whether the trajectory is acceptable, a reference boundary is needed. We introduce the *stability envelope*: a series of circular cross-sections at each time instant whose radii define the maximum acceptable phasor magnitude at that moment.

In the current implementation, the envelope is derived from a reference system—a known well-damped configuration. For the power system LFC model, this is a conventional synchronous-machine-dominated grid with inertia constant $H = 6$ s. For the RLC circuit, this is the Butterworth (maximally flat) response at $\zeta = 1/\sqrt{2}$.

The envelope defines a funnel-shaped surface in three-dimensional space. A stable system's helix collapses within this funnel. A system with degraded margins may overshoot the funnel during the initial transient but must return within it as time progresses. A system that persistently exceeds the envelope—particularly in the tail of the response—has insufficient damping.

5 Relationship to Classical Diagrams

The Dynamic Phasor Helix contains the information of the Bode diagram, the Nyquist diagram, and the step response as projections:

- **View from above** (collapse the time axis): the helix projects onto a spiral in the complex plane. For the step response, this is the phase portrait. For a frequency sweep, this reduces to the Nyquist plot.
- **View from the side** (collapse the complex plane to a single radial axis): the helix projects onto amplitude versus time—the classical step response envelope.
- **Read magnitude and phase at each height**: this recovers the information in the Bode diagram, though parameterized by time rather than frequency.

The connection between frequency-domain margins and time-domain behavior is made explicit through the pole locations. The dominant pole $s = \sigma \pm j\omega_d$ determines both:

- The **Bode phase margin**, which is related to the damping ratio $\zeta = -\sigma/\sqrt{\sigma^2 + \omega_d^2}$.
- The **helix decay rate**, which is directly $e^{\sigma t}$.

More phase margin $\Rightarrow$ more negative σ $\Rightarrow$ faster helix collapse $\Rightarrow$ tighter funnel. The geometric intuition is immediate.

6 Application: Power System Load Frequency Control

The single-area LFC model from Kundur [3], Chapters 11–12, provides the test case. The open-loop transfer function is:

$$L(s) = \left[\frac{1}{R} + \frac{K_i}{s}\right] \cdot \frac{1}{1 + sT_g} \cdot \frac{1}{1 + sT_t} \cdot \frac{1}{2Hs + D} \tag{5}$$

The physical interpretation of each parameter maps directly onto the helix behavior:

Table 1: LFC parameters and their effect on the helix

Parameter	Physical meaning	Effect on helix
H	Inertia constant [s]	Controls pole frequency; low H widens helix
D	Load damping [pu]	Dissipates energy; high D reduces spiral count
R	Droop [pu]	Proportional gain $= 1/R$; sets steady-state radius
T_g	Governor time constant [s]	Delays correction; increases overshoot
T_t	Turbine time constant [s]	Delays power delivery; extends oscillation
K_i	AGC integral gain	Eliminates steady-state offset; adds phase lag

The LFC system is structurally a PI controller (droop $= P$, AGC $= I$) acting on a plant with natural derivative-like behavior from inertia ($2Hs$ in the denominator). The helix makes this cascade visible: the initial overshoot and oscillations are governed by primary control (governor dynamics), the slow collapse of the residual offset is governed by secondary control (AGC integral), and the height at which the helix finally reaches zero radius indicates which control layer did the work.

Figure 1: Dynamic Impedance Phasor Helix for the single-area LFC model. Parameters: $H = 6.0$ s, $D = 1.0$ pu, $R = 0.05$, $T_g = 0.30$ s, $T_t = 0.5$ s, $K_i = 0.05$ with AGC enabled. The helix collapses within the green reference envelope ($H = 6$ s conventional system), indicating stable operation. The dominant complex poles at $s = -0.51 \pm 1.55j$ yield a damping ratio $\zeta = 0.311$ and oscillation frequency $\omega_d = 1.546$ rad/s. The phase portrait (floor plane) shows the spiral trajectory of Δf vs. $d\Delta f/dt$; the vertical axis is time.

7 Application: Series RLC Circuit

The series RLC circuit driven by a step voltage yields the transfer function for capacitor voltage:

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{6}$$

where $\omega_n = 1/\sqrt{LC}$ and $\zeta = (R/2)\sqrt{C/L}$.
The parameter mapping between the two domains is exact:

Table 2: Analogy between RLC circuit and power system LFC

RLC Circuit	Power System	Role
L (inductance)	$2H$ (inertia)	Resists changes in state variable
R (resistance)	D (damping)	Dissipates oscillation energy
$1/C$ (elastance)	$1/R$ (droop gain)	Restoring force
V_C (capacitor voltage)	Δf (frequency deviation)	State variable
V_{step} (applied voltage)	ΔP_{load} (disturbance)	Forcing function

The helix for both systems is constructed identically. Only the axis labels change. This universality confirms that the visualization is not domain-specific but applies to any second-order system with complex impedance.

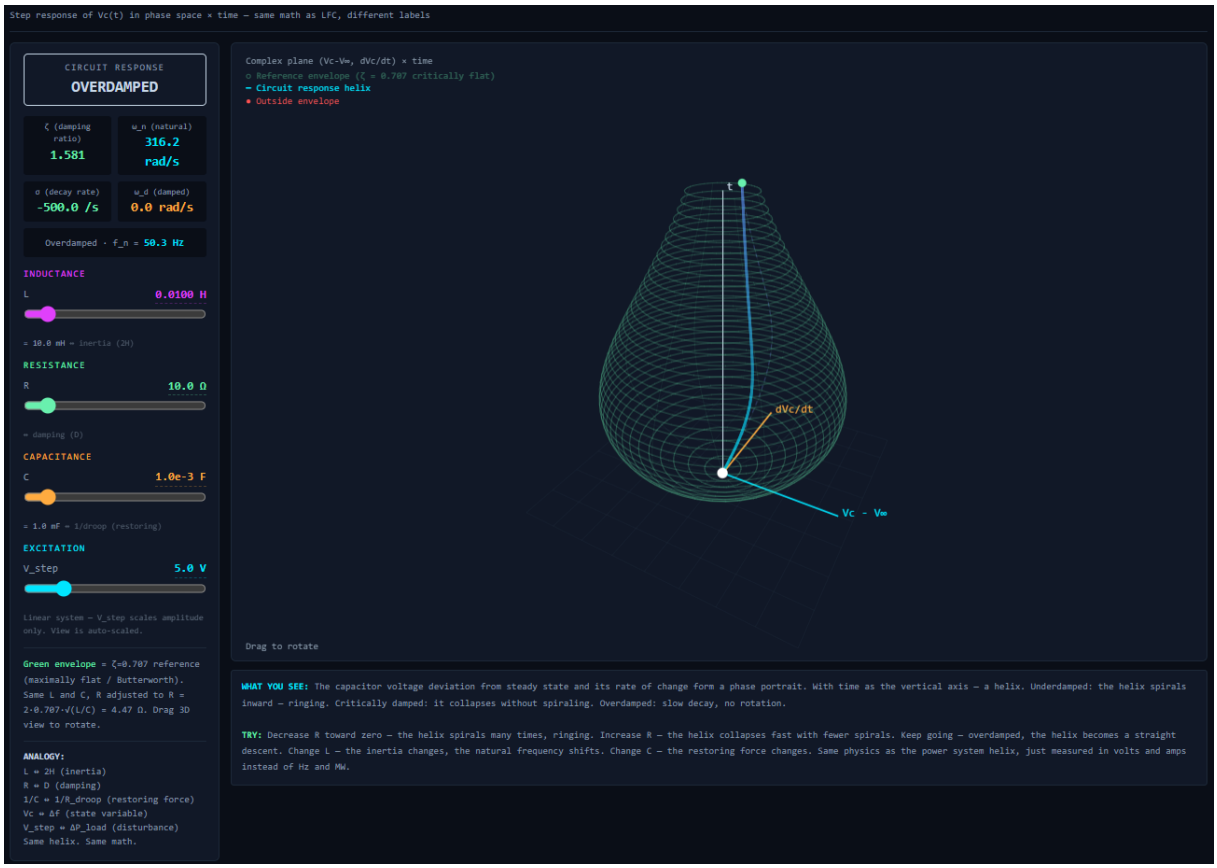

Figure 2: Dynamic Impedance Phasor Helix for a series RLC circuit. Parameters: $L = 10.0$ mH, $R = 10.0\ \Omega$, $C = 1.0$ mF, $V_{\text{step}} = 5.0$ V. This configuration is overdamped ($\zeta = 1.581$), producing a helix with no spiral—the trajectory descends directly without rotation. The natural frequency is $\omega_n = 316.2$ rad/s ($f_n = 50.3$ Hz) and the decay rate is $\sigma = -500.0$ /s. The green reference envelope corresponds to the Butterworth response ($\zeta = 1/\sqrt{2}$, $R = 4.47\ \Omega$). The mathematical structure is identical to the LFC model in Fig. 1; only the axis labels differ.

8 The Impedance Interpretation

The transfer function denominator in both cases is an impedance. For the LFC plant:

$$Z_{\text{plant}}(s) = 2Hs + D \tag{7}$$

This is structurally identical to a series RL impedance: $Z = sL + R$. The inertia constant H plays the role of inductance (opposing changes in the state variable), and D plays the role of resistance (dissipating energy).

The complete loop transfer function $L(s)$ is a ratio of complex impedances at each frequency. The phasor at any frequency ω represents the impedance relationship between the correction effort and the system response. The helix is therefore an *impedance trajectory through time*: it shows how the system's effective impedance to disturbances evolves after an event.

This impedance interpretation extends to multi-inverter grid stability, where the Nyquist criterion is applied to the ratio of inverter output impedance to grid impedance. The helix could, in principle, visualize these impedance interactions in real time using sampled measurement data.

9 Practical Implications: The Iberian Blackout

On 28 April 2025, the Iberian Peninsula experienced a complete power blackout—the most severe in Europe in over two decades. The ENTSO-E Expert Panel Final Report [4], published 20 March 2026, concluded that the blackout resulted from multiple interacting factors including gaps in voltage and reactive power control, differences in voltage regulation practices, and uneven stabilization capabilities.

At the time of the event, renewable sources accounted for 78% of electricity generation, with solar alone contributing nearly 60% [5]. The majority used grid-following inverters with limited frequency-responsive capability.

From the perspective of the helix framework:

- Reduced H (inertia): synchronous machines replaced by inverter-based resources.
- Reduced D (damping): frequency-responsive loads replaced by VSDs and power-electronic loads presenting unity power factor to the grid.
- Non-linear trip behavior: inverters disconnect at threshold voltages rather than degrading gradually, breaking the linearity assumption.
- Insufficient observability: real-time monitoring failed to detect the developing cascade.

The first two factors widen the helix beyond the reference envelope. The third breaks the model entirely—the helix cannot represent a binary disconnect event. The fourth means that even if the helix had been computed in real time, the input data was not available to do so.

This motivates the potential application of the helix as a real-time monitoring tool, driven by live measurement data from IEC 61850 sampled values, rather than as a design-time analysis tool driven by model parameters.

10 Limitations

Both models presented are linear, small-signal approximations valid near an operating point. They do not capture:

- Inverter trip thresholds and reconnection hysteresis.
- Governor deadbands and rate limiters.
- Hydro turbine inverse response (non-minimum phase zero).
- Saturation and current limiting in power electronics.
- Large-signal transients where linearization breaks down.

The helix shows what a linearized system predicts. Reality departs from it when nonlinearities dominate—and that departure is precisely where cascading failures begin.

11 Conclusion

The Dynamic Impedance Phasor Helix is not a new mathematical result. The underlying theory—Bode's gain-phase relationship, Nyquist's stability criterion, the connection between pole locations and transient response—is well established and dates to the 1930s and 1940s.

The contribution is representational: the observation that the Bode and Nyquist diagrams are complementary two-dimensional projections of a three-dimensional object, and that visualizing this object directly—a helix in phase-time space with a stability envelope—may provide a more

immediate understanding of how a system responds to disturbances. The helix makes explicit what the classical tools leave implicit: the temporal evolution of the impedance phasor and its relationship to the stability boundary.

The identical mathematical structure of the power system LFC model and the series RLC circuit confirms the generality of the approach. Any second-order system with complex impedance—electrical, mechanical, thermal, or otherwise—can be represented as a helix.

Whether this representation offers practical advantages over the classical tools for design, analysis, or real-time monitoring remains to be validated through broader use and feedback from the control systems community.

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"You can't control what you can't sample."

Repository: https://github.com/ninjas1337/Dynamic-Impedance-Phasor-Helix

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